# Pediatric Assessment and Emergencies

PAT, vital signs, airway infections, febrile seizures, dehydration, arrest, choking and abuse

**Medic Basics**
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## PEDIATRIC ASSESSMENT TRIANGLE (PAT) AND VITAL SIGNS

| PAT SIDE | LOOK FOR (HANDS OFF, FROM THE DOORWAY) |
|---|---|
| **Appearance** | Tone, interaction, consolability, gaze, cry (TICLS) |
| **Work of breathing** | Stridor, grunting, retractions, flaring, tripod |
| **Circulation to skin** | Pallor, mottling, cyanosis |

**EXAM TIP** Breathing alone = respiratory distress; plus appearance = respiratory failure; circulation = shock; appearance alone = brain or metabolic; all three = cardiopulmonary failure.

*NASEMSO 2022 (from PALS 2020 and Fleming 2011). Sources vary; trend.*

| AGE | HR AWAKE | RR |
|---|---|---|
| **Newborn** | 100–205 | 30–60 |
| **1–12 mo** | 100–190 | 30–60 |
| **1–2 yr** | 100–190 | 24–40 |
| **2–3 yr** | 98–140 | 24–40 |
| **3–4 yr** | 80–140 | 24–40 |
| **4–6 yr** | 75–140 | 22–34 |
| **6–10 yr** | 75–140 | 18–30 |
| **10–12 yr** | 75–118 | 18–30 |
| **Over 12 yr** | 60–100 | 15–20 |

Hypotension: SBP <60 in the first month, <70 at 1–12 months, <70 + (2 × age in years) at 1–10 years, <90 over 10. Normal SBP with poor perfusion = compensated shock; hypotension = decompensated, a late sign.

## RESPIRATORY DISTRESS VS FAILURE

- Distress: alert or anxious; retractions, flaring, grunting; fast heart rate
- Failure: listless; fewer retractions, fading breath sounds; pale, dusky or cyanotic; slowing heart rate

**SAFETY** Failure: BVM at 1 breath every 2–3 s. Heart rate below 60 with poor perfusion despite effective oxygenation and ventilation: start CPR.

## CROUP, EPIGLOTTITIS, BRONCHIOLITIS

| FEATURE | CROUP | EPIGLOTTITIS | BRONCHIOLITIS |
|---|---|---|---|
| **Cause** | Parainfluenza | Bacteria; Hib now rare | RSV mostly |
| **Age** | 6 mo–3 yr | Any; now mostly adults | Under 2 yr |
| **Hallmark** | Barky cough, stridor; worse at night | Sudden; drooling, muffled voice, tripod | Wheeze, crackles after a cold |
| **X-ray** | Steeple sign | Thumb sign | — |
| **Care** | Dexamethasone; neb epi if stridor at rest | Upright, calm; expert airway | Gentle suction, $O_2$, fluids |

- Croup, paramedic (NASEMSO): dexamethasone 0.6 mg/kg (max 16 mg); nebulized epinephrine 5 mg (5 mL of 1 mg/mL), may repeat in 20 min, per protocol
- Ask about choking: a foreign body can mimic croup
- Bronchiolitis: no routine albuterol or steroids

**SAFETY** Epiglottitis: no throat exam, IV attempt or lying flat. Keep the child upright and calm, give $O_2$ gently, be ready for a difficult airway.

## FEBRILE SEIZURES AND DEHYDRATION

- Age 6–60 months, ≥38 °C (100.4 °F), no CNS infection
- Simple: generalized, under 15 min, once in 24 hr. Complex: focal, 15 min or longer, or repeats within 24 hr
- About 30% recur; antipyretics only comfort and do not prevent seizures; no NSAIDs under 6 months
- Over 5 min or seizing on arrival: airway, $O_2$, glucose, benzodiazepine (paramedic: midazolam 0.2 mg/kg IM/IN, max 10 mg, per protocol)

**SAFETY** Under 6 months or over 5 years, or meningeal signs: not a simple febrile seizure; look for meningitis or another cause.

*CDC 2003; per protocol. Infants: sunken fontanelle.*

| DEGREE | SIGNS | FLUIDS |
|---|---|---|
| **Mild–moderate (3–9%)** | Fast pulse, thirst, dry mouth, fewer wet diapers | Oral rehydration 50–100 mL/kg over 2–4 hr |
| **Severe (over 9%)** | Lethargy, weak pulses, slow refill, little urine | 20 mL/kg LR or NS IV, repeat until perfusion recovers |

## CARDIAC ARREST AND CHOKING (2025)

*Exam-study doses; never above the adult dose. Follow your institution's or agency's protocols.*

| ITEM | INFANT AND CHILD |
|---|---|
| **Child age** | 1 yr to puberty (breasts, axillary hair); then adult BLS |
| **Depth, rate** | ≥⅓ of the chest: about 1.5 in (4 cm) infant, 2 in (5 cm) child; 100–120/min, full recoil |
| **Ratio** | 30:2 alone; 15:2 with 2 rescuers before puberty; advanced airway: continuous compressions, 1 breath every 2–3 s |
| **Infant hands** | Heel of 1 hand or 2 thumbs encircling; no 2-finger |
| **Defibrillator** | Manual if trained; else AED with pediatric attenuator (under 8 yr); else standard AED |
| **Shocks** | 2 J/kg, then 4; later ≥4, max 10 J/kg or adult dose |
| **Epinephrine** | 0.01 mg/kg IV/IO (0.1 mL/kg of 0.1 mg/mL), max 1 mg, every 3–5 min |
| **Antiarrhythmic** | Amiodarone 5 mg/kg (max 300 mg; up to 3 doses, repeats max 150 mg) or lidocaine 1 mg/kg |

**EXAM TIP** Epinephrine: nonshockable, as early as possible; shockable, after the second shock unless defibrillation is delayed.

- Severe choking: infant, 5 back blows then 5 chest thrusts (no abdominal thrusts); child, 5 back blows then 5 abdominal thrusts
- Unresponsive: start CPR (no pulse check); remove only a visible object, no blind finger sweeps

## CHILD ABUSE AND MANDATORY REPORTING

- Story that changes or does not fit; delayed care
- Under 4 yr: bruises on torso, ears, neck, frenulum, jaw angle, cheeks, eyelids or eye whites, or patterned; any bruise ≤4.99 months (TEN-4-FACESp)
- Immersion (stocking, glove) or cigarette burns; injuries of different ages
- Posterior or lateral rib or metaphyseal corner fractures; any fracture before walking
- Unexplained infant vomiting, lethargy, seizures, apnea or BRUE: think abusive head trauma
- Nurses must report in every state; EMS duties follow state law
- Suspicion is enough; do not investigate or ask leading questions
- Report to child protective services or police; a supervisor alone may not suffice
- Good-faith reports carry legal immunity
- Document findings and the child's own words

**SAFETY** Never confront a suspected abuser; get the child to safety and tell the receiving team.

**Sources** (20) are listed with links at medicbasics.com/sheets/pediatric-emergencies.html. For education and exam preparation only. Not medical advice; follow current guidelines and your institution's protocols. Medic Basics is an independent study aid not affiliated with any certifying body.